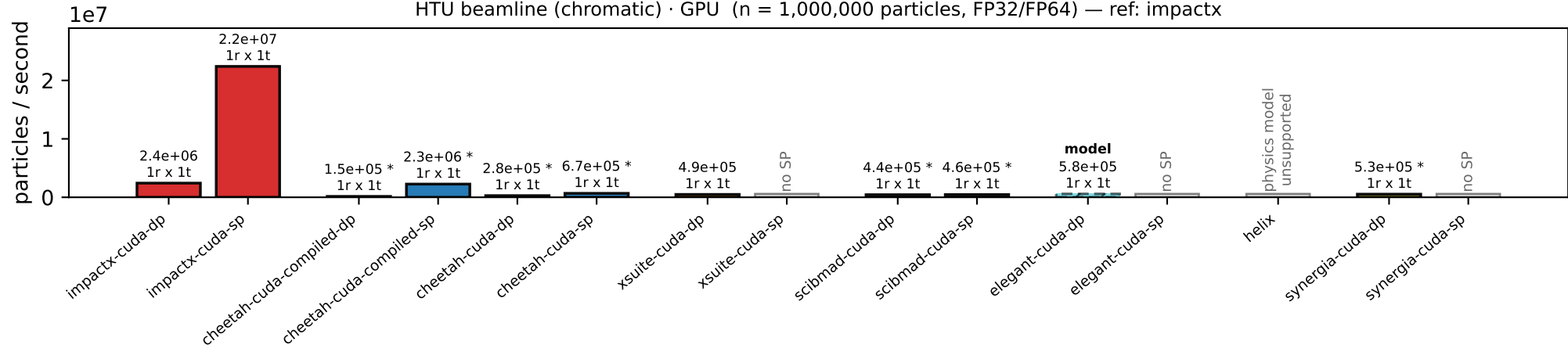


HTU beamline (chromatic) · GPU (n = 1,000,000 particles, FP32/FP64) — ref: impactx



lighter bar = fast-math (relaxed FP), drawn behind its IEEE bar

* Cheetah/SciBmad/Bmad/Synergia have no chromatic-paraxial model here; they run the costlier exact map (agrees within tol)

versions: impactx 26.08 · cheetah 0.8.2 · xsuite 0.53.4 · scibmad v0.4.1 · elegant elegant-2026.3.0-52-g8b7ea15 · helix v1.9.0 · synergia v2024.04-209-g795b99f39